Feldstein also said that emerging markets could not rely on the IMF or other international organizations; emerging markets must save on their own. They did just as he instructed and beefed up their sovereign savings during the 2000s.

In the argot of economists, SWF savings are needed for *precautionary* reasons and are a form of *self-insurance*.[7] But perhaps the AUM in SWFs are too high; Lee (2004) reports reserves of 17% of GDP, on average, for emerging market countries. Just as miserly Ebenezer Scrooge initially misses out on the joys of Christmas in Charles Dickens' novel *A Christmas Carol*, excessive saving by a country reduces consumption. Countries could use this capital more productively elsewhere in the economy, and offer better social insurance, health, or education programs (making life more enjoyable for Scrooge's poor clerk, Bob Cratchit, and his disabled son, Tiny Tim). The optimal level of sovereign savings is a balancing act between the opportunity costs of consumption and investment, and the value of self-insurance.

2.3. DUTCH DISEASE

Timor-Leste created its SWF not out of precautionary savings motives; it was to avoid the *Dutch disease*.

The Dutch disease (or *resource curse*) was a term first used by the *Economist* in 1977 for the shrinking of the manufacturing sector in the Netherlands after natural gas was discovered in the previous decade. The same effect occurred in Britain in the 1970s when oil was found under the North Sea. Once a country finds natural resources, manufacturing and other traded sectors decline while real exchange rates appreciate, causing the country's traded sectors to be less competitive. Resource bonanzas also increase corruption and wasteful government spending.[8]

Nigeria is perhaps the worst example of the Dutch disease.[9] Oil prices rose dramatically from the 1960s to the 2000s, but Nigeria has always been among the twenty poorest countries in the world. Between 1970 and 2000, the proportion of the population surviving on less than $1 per day increased from 25% to 70%. A series of military dictatorships have plundered the oil money.

[7] In Jeanne and Rancière's (2006) language, the SWF can sustain demand in times of "sudden stops." Heller (1966) is the first to develop the precautionary savings story for sovereign reserves. The Greenspan–Guidotti rule, which is named after a former chairman of the Federal Reserve Board and a former deputy minister of finance of Argentina, recommends a country hold reserves equal to its short-term external debt. An alternative, *new mercantilist* explanation for the rise of sovereign wealth is Dooley, Folkerts-Landau, and Garber (2005). They argue that the increased sovereign wealth arises as a product of some countries wanting to fix exchange rates. Related work on international savings (or equivalent dis-savings by some countries) is by Gourinchas and Rey (2007).

[8] See van der Ploeg (2011). A model of the Dutch disease, and the use of a SWF to counter it, is Collier et al. (2010).

[9] As recounted here by Sala-i-Martin and Subramanian (2003).